Physics 5313: Electromagnetic Theory 2 (Fall 2026)

Lecture 13: Energy Attenuation in Waveguides 2 and Resonant Cavity

Instructor: Zihan Wang

General form

$$P = \frac{1}{2\sqrt{\mu\epsilon}} \left(\frac{\omega}{\omega_\lambda}\right)^2 \left(1 - \frac{\omega_\lambda^2}{\omega^2}\right)^{1/2} \left\{\begin{matrix}\epsilon \\ \mu\end{matrix}\right\} \int_A \psi^* \psi \, da$$

$$-\frac{dP}{dz} = \frac{1}{2\sigma\delta} \left(\frac{\omega}{\omega_\lambda}\right)^2 \oint_C \left\{\begin{matrix}\frac{1}{\mu^2\omega_\lambda^2} \left|\frac{\partial\psi}{\partial n}\right|^2 \\ \frac{1}{\mu\epsilon\omega_\lambda^2}\left(1 - \frac{\omega_\lambda^2}{\omega^2}\right)|\mathbf{n} \times \nabla_t\psi|^2 + \frac{\omega_\lambda^2}{\omega^2}|\psi|^2\end{matrix}\right\} dl$$

$$\beta = -\frac{1}{2} \frac{dP/dz}{P} \qquad \beta(\omega)$$

$\nabla_t^2 \psi + \gamma_\lambda^2 \psi = 0$ $\quad$ L is the spatial scale of ψ.

$\nabla_t^2 \psi \approx \frac{\psi}{L^2} \Rightarrow L \sim \frac{1}{\gamma_\lambda}$

General form

$$\left\langle \left|\frac{\partial \psi}{\partial n}\right|^2 \right\rangle \sim \left\langle \left|\vec{n} \times \nabla_t \psi\right|^2 \right\rangle \sim \gamma_\lambda^2 \left\langle |\psi|^2 \right\rangle$$

$$\langle x \rangle = \frac{1}{C} \int x \, dl$$

C is the total length of dl

$$\int \left|\frac{\partial \psi}{\partial n}\right|^2 dl \sim \gamma_\lambda^2 \int |\psi|^2 dl = \gamma_\lambda^2 \frac{C}{A} \xi_\lambda \int |\psi|^2 da$$

A is the area of the cross section

TM: $$\beta_\lambda = \frac{1}{2} \frac{\sqrt{\mu \epsilon}}{\sigma \delta} \frac{1}{\left(1 - \frac{w_\lambda^2}{w^2}\right)^{1/2}} \frac{\frac{1}{\mu^2 w_\lambda^2} \gamma_\lambda^2 \frac{C}{A} \xi_\lambda}{2}$$

$$= \frac{C}{2A} \frac{\sqrt{\epsilon/\mu}}{\sigma \delta_\lambda} \frac{\left(w/w_\lambda\right)^{1/2}}{\left(1 - w_\lambda^2/w^2\right)^{1/2}} \xi_\lambda \qquad \delta_\lambda = \left(\frac{2}{\mu_c w_\lambda \sigma}\right)^{1/2}$$

General form

$$\frac{d\beta_\lambda}{dw} = 0 \qquad \frac{d}{dw}\left(\frac{\left(\frac{w}{w_\lambda}\right)^{1/2}}{\left(1 - w_\lambda^2/w^2\right)^{1/2}} \right) = 0$$

$$\Rightarrow \quad w = \sqrt{3}\, w_\lambda$$

General form

$$\text{TE}: \quad \int |\vec{n}\times\nabla_t\psi|^2\,dL = \gamma_\lambda^2\frac{C}{A}\xi_\lambda\int|\psi|^2\,da \qquad ①$$

$$\frac{w_\lambda^2}{w^2}\int\left(|\psi|^2-\frac{1}{\gamma_\lambda^2}|\vec{n}\times\nabla_t\psi|^2\right)dL \qquad ②+③$$

$$=\frac{w_\lambda^2}{w^2}\eta_\lambda\frac{C}{A}\int|\psi|^2\,da$$

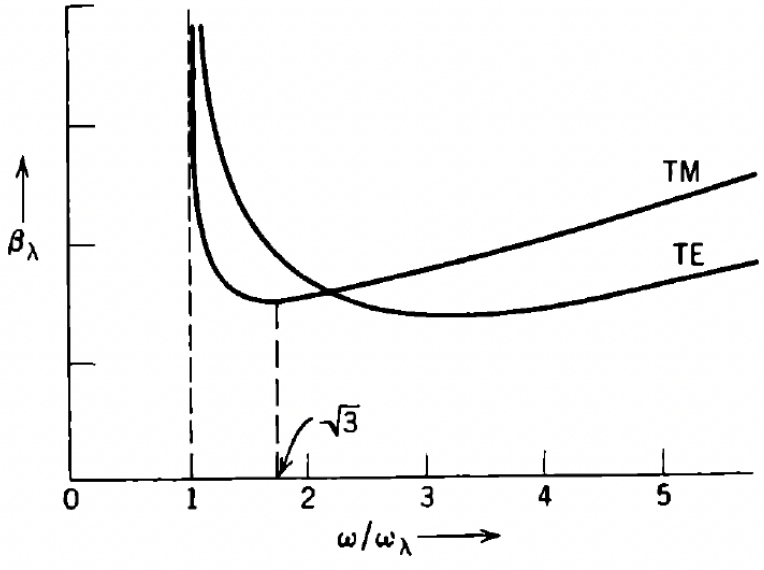
$$①+②+③\quad \beta_\lambda=\sqrt{\frac{\varepsilon}{\mu}}\,\frac{1}{b\delta\lambda}\left(\frac{C}{2A}\right)\frac{(w/w_\lambda)^{1/2}}{(1-w_\lambda^2/w^2)^{1/2}}\left(\xi_\lambda+\frac{w_\lambda^2}{w^2}\eta_\lambda\right)$$

$$\frac{d\beta_\lambda}{dw}=0 \qquad \text{minimum depends on the } \xi_\lambda,\ \eta_\lambda$$

General form

TE_{10} mode :

$\xi_{10} = \frac{a}{a+b}$ $\eta_{10} = \frac{2b}{a+b}$



M2: Perturbation of Boundary Conditions

- What is α_λ?

$$k = k_\lambda + \alpha_\lambda + i\beta_\lambda$$

Perfect : $\vec{E}_{//} = 0$

Good : $\vec{E}_{//} = \sqrt{\frac{2\mu_c w}{2b}}\,(1-i)(\vec{n}\times\vec{H}_{//})$

$(\nabla_t^2 + \gamma^2)\psi = 0$

$\psi = \psi_0 + \psi'$ $\quad \psi$ is solution in good.

ψ_0 , , , , perfect.

$\psi' \ll \psi_0$

M2: Perturbation of Boundary Conditions

- Boundary condition

$(\nabla_\phi^2 + \gamma_0^2)\,\psi_0 = 0 \qquad \psi_0|_S = 0 \qquad$ Perfect

$(\nabla_\phi^2 + \gamma^2)\,\psi = 0 \qquad \psi|_S \neq 0 \qquad$ Good

M2: Perturbation of Boundary Conditions

TM mode : $H_z = 0$

$$\vec{E}_{//} = \sqrt{\frac{\omega\mu_c}{2\sigma}}\,(1-i)\,(\hat{n}\times\vec{H}_{//}) = \sqrt{\frac{\omega\mu_c}{2\sigma}}\,(1-i)\,\hat{n}\times\vec{H}_t)$$

$\hat{n}\times\vec{H}_{//}$ is in $\hat{z}$ direction

$\vec{E}_{//}$ is , ,

$$E_z = \vec{E}_{//}\cdot\hat{z} = \sqrt{\frac{\omega\mu_c}{2\sigma}}\,(1-i)\,(\hat{n}\times\vec{H}_t)\cdot\hat{z}$$

$$= \sqrt{\frac{\omega\mu_c}{2\sigma}}\,(1-i)\left[\hat{n}\times\left(\frac{i}{\delta_0^2}\zeta\omega\hat{z}\times\nabla_t E_z\right)\right]\cdot\hat{z}$$

M2: Perturbation of Boundary Conditions

$$\bar{E}_z = \sqrt{\frac{\mu_c \omega}{2\sigma}}(1-i)\left[\vec{n}\times\left(\frac{i}{\gamma_0^2}\zeta\omega\hat{z}\times\nabla_0\bar{E}_z\right)\right]\cdot\hat{z}$$

$$= \sqrt{\frac{\mu_c\omega}{2\sigma}}(1-i)\frac{i}{\gamma_0^2}\zeta\omega\frac{\partial\bar{E}_z}{\partial n}$$

$$= \sqrt{\frac{\mu_c\omega}{2\sigma}}(1+i)\frac{\omega}{\omega_0^2\mu}\frac{\partial\bar{E}_z}{\partial n} \qquad \delta = \sqrt{\frac{2}{\mu_c\omega\sigma}}$$

$$= (1+i)\frac{\mu_c\delta}{2\mu}$$

M2: Perturbation of Boundary Conditions

$$\bar{q}_z = (1+i)\frac{\kappa_c \delta}{2\kappa}\frac{w^2}{w_0^2}\frac{\partial \bar{q}_z}{\partial n}$$

Assume $$\frac{\partial \bar{q}_z}{\partial n} \approx \frac{\partial \bar{q}_{z0}}{\partial n}$$

$$\bar{q}_z = (1+i)\frac{\kappa_c \delta}{2\kappa}\frac{w^2}{w_0^2}\frac{\partial \bar{q}_{z0}}{\partial n}$$

introduce $$f = (1+i)\frac{\kappa_c \delta}{2\kappa}\frac{w^2}{w_0^2}$$

$$\bar{q}_z = f\frac{\partial \bar{q}_{z0}}{\partial n}$$

M2: Perturbation of Boundary Conditions

$$\begin{cases} (\nabla_t^2 + k^2)\psi = 0 \\ \psi = f \dfrac{\partial \psi_0}{\partial n} \quad \text{on } S \end{cases}$$

Green's Theorem

$$\int_A (\phi \nabla_t^2 \psi - \psi \nabla_t^2 \phi)\, da = \oint (\psi \frac{\partial \phi}{\partial n} - \phi \frac{\partial \psi}{\partial n})\, dl$$

$\psi = \psi \quad \phi = \psi_0^*$

M2: Perturbation of Boundary Conditions

$$\int_A (\psi_0^* \nabla_\phi^2 \psi - \psi \nabla_\phi^2 \psi_0^*)\, da = \oint \left(\psi \frac{\partial \psi_0^*}{\partial n} - \psi_0^* \frac{\partial \psi}{\partial n}\right) dl$$

BC. $\psi_0 = 0 \Rightarrow \psi_0^* = 0$

$$\psi = f \frac{\partial \psi_0}{\partial n}$$

Laplace Eq: $\nabla_\phi^2 \psi = -\gamma^2 \psi$

$$\nabla_\phi^2 \psi_0 = -\gamma_0^2 \psi_0$$

M2: Perturbation of Boundary Conditions

$$\int_A (\psi_0^* \nabla_0^2 \psi - \psi \nabla_0^2 \psi_0^*)\, da = \oint \left(\psi \frac{\partial \psi_0^*}{\partial n} - \psi_0^* \frac{\partial \psi}{\partial n}\right) dl$$

$$\int_A \left(-\gamma^2 \psi_0^* \psi + \gamma_0^2 \psi \psi_0^*\right) da = \oint f \frac{\partial \psi_0}{\partial n} \frac{\partial \psi_0^*}{\partial n}\, dl$$

$$(\gamma_0^2 - \gamma^2) \int \psi_0^* \psi \, da = f \oint \left|\frac{\partial \psi_0}{\partial n}\right|^2 dl$$

$$\psi \approx \psi_0$$

$$(\gamma_0^2 - \gamma^2) \int |\psi_0|^2 da = f \oint \left|\frac{\partial \psi_0}{\partial n}\right|^2 dl$$

M2: Perturbation of Boundary Conditions

$$\beta^2 + k^2 = \mu\varepsilon\omega^2$$

$$k^2 - k_0^2 = f\,\frac{\oint \left|\frac{\partial \psi_0}{\partial n}\right|^2 dL}{\int |\psi_0|^2\, da}$$

$$\frac{k^2 - k_0^2}{f} = \beta = \frac{1}{2}\,\frac{\frac{1}{2\beta S}\left(\frac{w}{w_0}\right)^2 \frac{1}{\pi^2 w_0^2}\oint \left|\frac{\partial \psi_0}{\partial n}\right|^2 dL}{\frac{1}{2\sqrt{\pi\varepsilon}}\left(\frac{w}{w_0}\right)^2\left(1 - \frac{w_0^2}{w^2}\right)^{1/2}\varepsilon \int |\psi_0|^2\, da}$$

M2: Perturbation of Boundary Conditions

$$\beta = \frac{1}{2(1+i)} \frac{k^2 - k_0^2}{k_0}$$

$$k^2 = k_0^2 + 2(1+i) k_0 \beta = k_0^2 \left(1 + \frac{2(1+i)\beta}{k_0}\right)$$

$$k = k_0 \left(1 + \frac{2(1+i)\beta}{k_0}\right)^{1/2}$$

Assume $k_0 >> \beta \Rightarrow k = k_0 + (1+i)\beta = k_0 + \beta + i\beta$

$$k = k_0 + \alpha + i\beta \qquad \alpha = \beta$$

Microwave in copper WG. $\beta \approx 10^{-4} \frac{\omega_\lambda}{c}$ $k = \frac{\omega}{c}$